

Using Data and Information

All projects will require data and information, and in a project environment, being able to access, retrieve, utilise and safely store data and information will be vital to its success.

Data and information are often used interchangeably, but there is an important distinction between them. The CMI defines data as simply ‘facts and figures’. They are the smallest, most basic units of evidence, which is why they’re often described as ‘raw.’ By contrast, ‘information’ is defined as ‘data that has been put into context, analysed and interpreted’¹. In other words, information is the result of organising and analysing data. It’s also the most useful for making decisions.

This information provides some tips on the best way to access and use data and information, along with some of the challenges you may face. You will also find a companion sheet: ‘Analysing and Summarising Quantitative Data’ which offers guidance on data analysis.

Sources of Data and Information

To access the data and information most useful for your project, you need to fully understand **where** and in what **form** they appear. Both data and information may be categorised according to these two features.

You may already be familiar with the terms ‘primary’ and ‘secondary’ - these distinguish data and information based on from **where** they originate, or their source. *Primary data* is that which you or your project team have gathered for a particular project. *Secondary data*, on the other hand, has been compiled by someone else for their own purposes. However, it may still be quite useful to you. For example, secondary information could include a report or guidance published by the Association of Project Management.

The **form** of data and information will be either ‘quantitative’ or ‘qualitative’. *Quantitative data* is expressed in numbers - examples include salaries, timelines and profit². These are characteristics that can be measured, or ‘quantified.’ Quantitative data could provide you with breadth and detail for your project. *Qualitative data*, by contrast, cover results that can’t be easily measured or counted. Qualitative data is subjective - it involves judgements, such as observations and opinions. It will offer depth to information by helping to explain phenomena identified by quantitative data. For example, a fall in sales can be measured using quantitative data but to establish why they dropped may require some explanation, gained qualitatively.

Both these forms of data and information may be useful to you in the development of your project.

¹ CMI (2019) ‘CMI Checklists - 166 Knowledge Management’ *Management Direct*

² Payne, E. & Whittaker, L. (2006) *Developing Essential Study Skills*, Pearson, Harlow

Data and Information Quality

To support your project development most effectively, you will need to source the best quality data and information you can find. While there isn't a single, universally agreed set of quality criteria, the most commonly cited measures are: relevance, accuracy, reliability, completeness and currency. Some of these criteria focus more on data and some concentrate on information.

We'll take a closer look at our five criteria now.³

- **Relevance**

Relevance simply means that the data or information meets your needs and serves your purposes. This means that it must be directly related to your project goals in terms of its setting, population or time period.

- **Accuracy**

Accuracy is the degree to which your data and information reflects reality. However, it's important to note that there is a difference between data accuracy and information accuracy.

The accuracy of data will depend on how well it has been recorded and if it is recorded in the agreed format. For example, on a spreadsheet, if some dates have been entered as YY/MM while others follow MM/YY format, some data entries won't be valid, and therefore the data as a whole will not be accurate. Data must also be unique. This means that the same fact should not be counted twice. For example, a person's name may be entered under both Robert Smith and Bob Smith in the same spreadsheet, thereby affecting the accuracy of the data.

By contrast, information accuracy depends on how competently data has been analysed. For example, it would be compromised if those analysing the information aren't proficient at statistics, or if they have used analytical software - such as Excel - incorrectly, leading to inaccurate results.

Accuracy can also be affected by the degree to which our sources are trustworthy or representative, but these elements are significant enough that they require separate consideration - under the criterion of reliability.

- **Reliability**

Reliability is closely related to accuracy. In fact, unreliable data - such as data that have been collected by a biased individual - are often also inaccurate. However, reliability merits its own category because data and information may be technically accurate, yet still not be reliable. This is because the circumstances that produce data and information can change.

Data collected at a certain time may not reflect the reality across a longer period. For example, a project that uses data on the characteristics of holidaymakers would be limited if it was gathered

³ Office for National Statistics (2019) 'Quality in official statistics - quality defined' available at <https://www.ons.gov.uk/methodology/methodologytopicsandstatisticalconcepts/qualityinofficialstatistics/qualitydefined>

only during the school term. It wouldn't reflect the diversity of holidaymakers - such as families travelling during school holidays. It would be accurate in terms of the time in question, but it wouldn't be reliable because of its failure to represent the bigger picture.

The individuals involved in the collection of data may go about it differently and therefore produce different results. This also applies to information, because two people may analyse data in a very different way. The test for whether data is reliable is that it would reflect the same findings if it was collected again. And for information, it's whether another analysis would produce the same result.

So, to assess reliability, you need to consider if the findings would be the same if a different entity collected or analysed the data. For example, does the individual or institution have a motive which might see them - consciously or unconsciously - wish for the data to show a specific outcome? In other words, are they biased? Could you or your project team attenuate any bias or concerns?

- **Completeness**

Completeness concerns how much data or information you possess in relation to your project needs. Do you have enough for a meaningful understanding of the situation? While the answer to this question is mostly subjective, there are some key considerations that can help you decide. In the case of primary data, the size of your sample can determine how complete it is. This is the amount of data you have collected on the population you are studying - whether that is employees, clients or customers.

You don't need to have a 100 percent participation rate for the data to be complete. For example, you don't need everyone to reply to a survey; it's a matter of getting enough data to be representative of the population. What that means will depend on its size - the smaller the population, the higher the response rate needs to be. In the case of secondary data, you'll need to seek out the original research and find out the sample size.

The completeness of information can be more complex to assess, but the same principle applies: do you have a sufficient amount of information for your needs?

Considering both the breadth and depth of the information can be useful. For example, if your project involved selecting a new training provider, you would need to look at a range of criteria. These could include the quality of learning materials or learner feedback scores. The information would also have to be of a certain depth. In the case of the training provider, you would probably want to view multiple learning materials relating to one topic.

Completeness contributes directly to the reliability of data and information, because when data and information is incomplete, it may present a one-sided or misleading view. Let's say you've surveyed the employees of one department for their views on a possible office relocation. A response rate of 5% will not provide you with a meaningful understanding of the department as a whole, especially if the responses come predominantly from one group, such as managers.

- **Currency**

Currency relates to whether the data or information is sufficiently up-to-date. This will vary according to your particular context. For an ambulance service, a current statistic might be as recent as the last few days. In an accountancy firm, current might refer to the most recent quarter. In a farming context, it might apply to a whole season.

As with completeness, the insufficient currency of data and information can affect reliability. For the ambulance service, data on emergency callouts that isn't current enough may be unreliable, as they represent circumstances that have changed since the figures were collected.

Understanding how all these criteria affect the quality of your sources will help you to use them effectively in the development and delivery of your project.

So far, we have focused on data and information that you will access to support delivery of your project, but you will also be creating it too. In this case, the same quality criteria will apply.

Data Management and Security

Whether you have collected data and information from elsewhere or created it yourself, you will need to manage it, keep it safe, and make it easy to retrieve.

Although much of the data and information we use is digital, depending on the nature of your project, you may have some documents in their original paper form. Paper is particularly useful for longer-term storage as some digital formats can become obsolete over time. While a physical object may seem at higher risk of theft, it's also easier to protect, as paper copies typically cannot be accessed by those outside the organisation. However, a real problem with physical copies is that they can be destroyed by physical means - typically by fire or flood. Paper takes up space, so you'll need a good filing system.

In contrast, digital data and information offers quick and easy access, no matter where the document is located. This has been made easier with cloud technology, which makes documents accessible from mobile devices across the globe, providing huge flexibility. This can be very useful if you are part of a project team working across different locations.

Digital information can improve safety and security because documents are 'hidden', and protected by passwords, encryption and backup drives. However, it can also be more vulnerable to fraud than their hard copy equivalents. For example, cybercrime, such as phishing - where sensitive information is fraudulently accessed - is now widespread.

For greater security, it's best to store data and information in multiple formats; backup methods should be of prime importance to anyone responsible for data storage.

Data and Information Challenges

The sheer quantity of data and information that is processed in a day on even a moderate sized project can be intimidating. Three common challenges you may face are detailed below.

The first major challenge is 'information overload'. This happens when the volume of data and information confronting us exceeds our capacity to process it. The problem has worsened, however, because today, data and information exists in multiple forms and across multiple platforms, making its handling and analysis more complex.⁴ Data overload can be managed through filtering. This might include data from less trustworthy sources, or older information which may not provide an up-to-date picture.

A second challenge that you may face relates to the quality of your data and information. We have already looked at five criteria you can apply; however, the overall quality test should be whether it's fit for its intended uses, or in other words, supports your project delivery.

Our third challenge relates to access. You may face physical challenges - for example, you could be charged a hefty fee to download a commercial report, or it might only be available to subscribed members. You also need to think about readability, where the sheer volume, density or structure, may make some data and information difficult to comprehend.

Next time you are working with data and information as part of your project, think about how they can be used to best effect.

⁴ <https://www.ibmbigdatahub.com/blog/3-top-data-challenges-and-how-firms-solved-them>